

GOVERNMENT DATA INTELLIGENCE & ECONOMETRIC MODELING

India Tourism Analysis & Pipeline

Granular Post-COVID Dynamics, Hospitality Capacity & Aviation Load Modeling (2022–2026)

Author: Thillai valavan A S

Primary Sources: Ministry of Tourism (MoT), NIDHI Portal, ASI, BOI, DGCA / AAI

Executive Overview & Core Analytical Scope

474

Districts Tracked

Comprehensive footfall across all 28 states & 8 UTs (2022–2026)

1,650

Hospitality Units

NIDHI verified star hotels, resorts, lodges & homestays

1,200

ASI Monuments

Centrally protected national heritage monuments & UNESCO sites

40

Aviation Gateways

Monthly passenger traffic & flight movements across 60 months

Key Methodological & Econometric Pillars:

• End-to-End Clean Architecture:

Standardized repository layout separating untouched public datasets (data/raw/) from typed, imputed, analysis-ready datasets (data/cleaned/).

• Domain-Aware Imputation:

Eliminated statistical distortions using hierarchical modal pincodes, category-bounded medians, and structural zero logic for unticketed heritage sites

• Relational Master Integration:

Synthesized 'unified_district_master_2026.csv' linking district footfalls with hotel room capacity and UNESCO sites for infrastructure stress analysis

• Actionable Econometric Insights:

Identified high-velocity spenders (Oceania \$257.7/day), key inbound foreign destinations (Ernakulam 20.08% foreign share), and airport interconnectivity

Data Engineering & Domain-Aware Imputation Pipeline

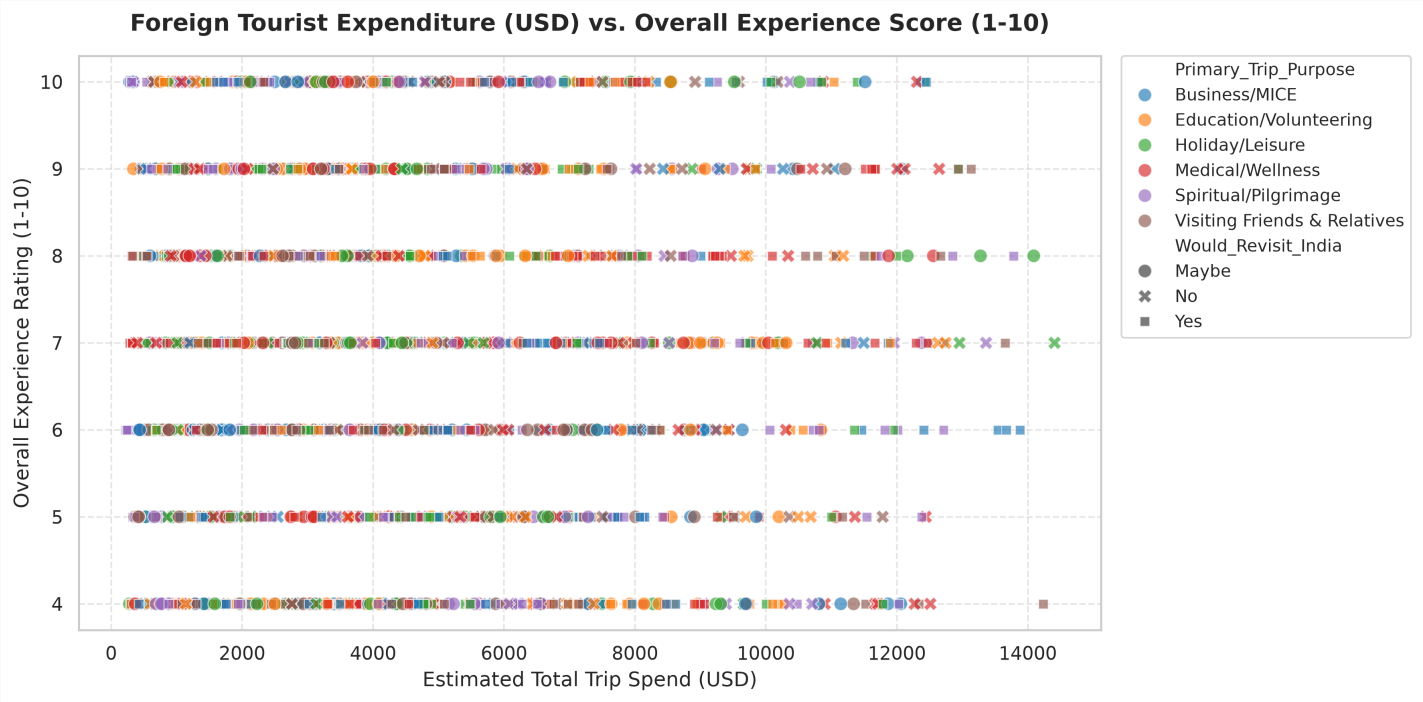
Data Type Optimization & Memory Efficiency

- ✓ **Nullable Int64 Casting**
Replaced standard float casting with pandas nullable Int64 on count columns with missing values (e.g. Registered_Hotels_Count), preserving integer integrity (preventing overflow).
- ✓ **Ordered Month Categoricals**
Cast survey and airport months to ordered categorical series, enabling valid cyclical and seasonal chronological indexing.
- ✓ **Categorical Reductions**
Cast high-cardinality repetitive strings (State_UT, Star_Classification, Architectural_Type) to categorical, reducing memory usage by up to 16x.
- ✓ **Standardized Timestamps**
Synthesized proper Year-Month datetime index across monthly aviation movements for seamless time-series aggregation.

Domain-Aware Missing Value Imputation

- ✓ **Hierarchical Pincode Mode**
Imputed missing establishment pincodes using the statistical mode within the exact district, falling back to 'Unknown' only if the whole district had no valid codes.
- ✓ **Explicit 'Unclassified' Star Tier**
Introduced explicit category token rather than inventing unearned star tiers or discarding real registered hospitality businesses.
- ✓ **Structural Zero Logic**
Assigned ■0 entry fees to non-ticketed heritage monuments and 0 international movements to purely domestic regional airfields.
- ✓ **Conditional Group Medians**
Imputed missing hotel room counts and tourist survey expenditure by (State, Category) and (Region, Purpose) clusters.

Foreign Inbound Spending Velocity & Visitor Experience



Econometric Insights:

- Daily Spending Velocity**

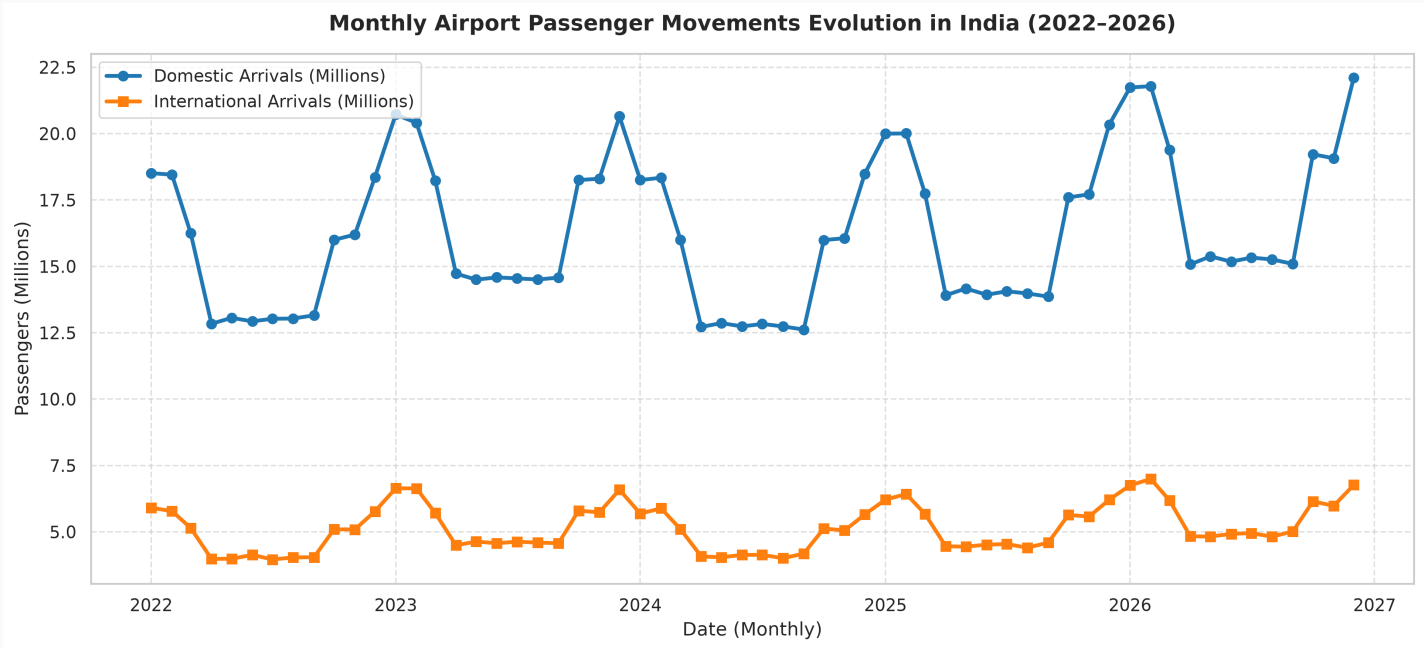
Visitors from Oceania lead with \$257.70/day, followed by Africa (\$246.70/day) and the Americas (\$224.50/day).
- Spend vs. Satisfaction**

Travelers with expenditure > \$6,000 USD report consistent ratings between 8 and 10 across holiday and business trips.
- Net Promoter Sentiment**

31.9% of South Asian visitors and 28.7% of American tourists classified as active Promoters.
- Extended Stay Dwell Time**

Average length of stay spans 20 to 26 nights across primary long-haul origin regions.

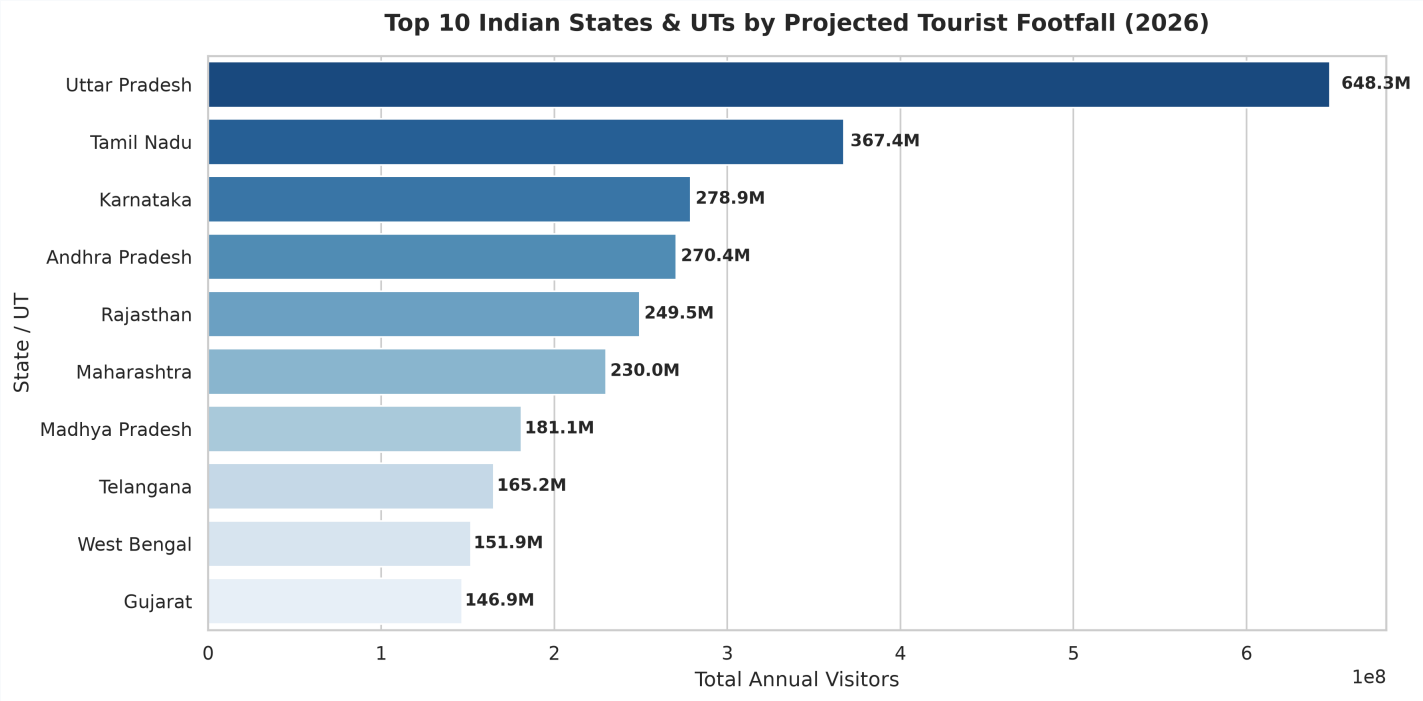
Aviation Passenger Movements & Airport Gateway Loads



Aviation Dynamics (2022–2026):

- **Multi-Year Recovery**
Continuous upward trajectory across monthly domestic and international movements from 2022 through 2026.
- **Domestic Dominance**
Domestic traffic represents 75-80% of aggregate passenger volumes across the national airport grid.
- **Volume Gateways**
Delhi (DEL) leads with 95.7M annual passengers in 2026, followed by Mumbai (BOM) at 80.3M and Bengaluru (57.8M).
- **International Exposure**
Cochin (COK) (48.4%) and Thiruvananthapuram (TRV) (46.0%) exhibit exceptionally high international dependency.

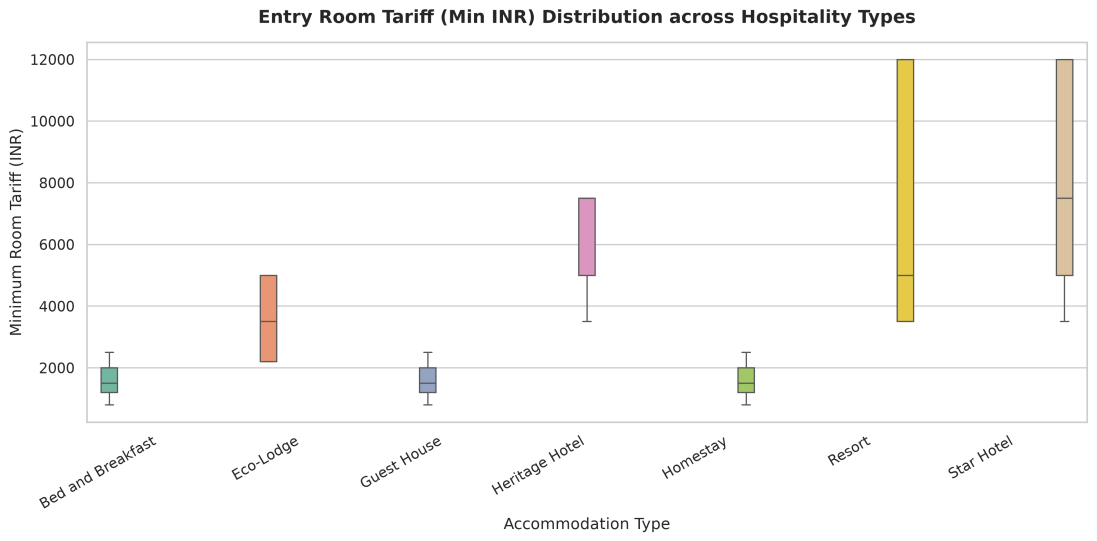
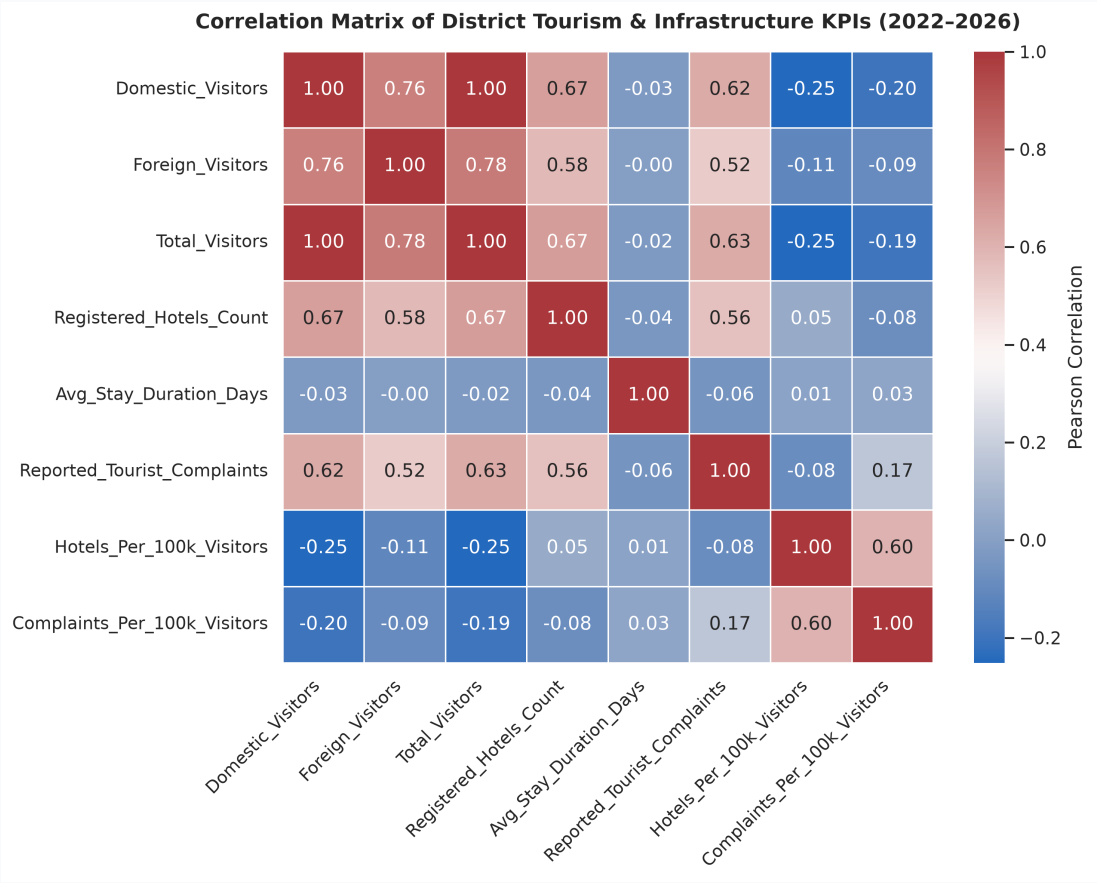
Top 10 Indian States & UTs by Projected Tourist Footfall (2026)



State Footfall Findings:

- Domestic Engine**
Southern and Western states continue to capture massive domestic volumes driven by pilgrimage and business corridors.
- Foreign Clusters**
Kerala, Tamil Nadu, West Bengal, and Rajasthan demonstrate the highest foreign-to-domestic capture ratios.
- Inbound Destinations**
Ernakulam (20.08% foreign share) and Puri (14.99%) record the highest inbound density at the district level.
- Infrastructure Stress**
Certain pilgrimage districts exhibit tight hotel-room to visitor ratios, highlighting immediate investment opportunity.

District Infrastructure Correlation & Hospitality Tariff Segments



Strategic Policy Recommendations & Next Steps

1. Inbound Density Support for High-Foreign Share Districts

Direct targeted multi-lingual desks, emergency tourist police outposts, and seamless digital international payment rails to high foreign-share hubs (Ernakulam 20.08%, Puri 14.99%).

2. Targeted Homestay & Bed-and-Breakfast Quality Auditing

Bridge the hospitality audit score gap (homestays average 75.7 vs eco-lodges 79.7) through decentralized district-level certification incentives under NIDHI.

3. Aviation Capacity Expansion at Diaspora Gateways

Prioritize terminal expansion and international wide-body parking bays at high-exposure regional gateways such as Cochin (48.4% intl) and Thiruvananthapuram (46.0% intl).

4. Eco-Tourism & Nature Circuit Promotion

Capitalize on highest foreign dwell time (3.0 days) and high spending velocity by funding low-impact green hospitality infrastructure in nature & wildlife corridors.